Find the common fields, then index them

Read three kinds of field, classify the values, and write one entity index for hunting.

Every table in the lake

Microsoft tables, CommonSecurityLog, Syslog, custom _CL tables. A Spark notebook lists and samples them without the interactive row cap.

Typed columns

Read the table schema (`getschema`, `df.schema`)

SigninLogs.UserPrincipalName

JSON and dynamic columns

Pull keys from sampled rows (`bag_keys`)

AdditionalFields, RawEventData, Properties

Key=value and raw text

`parse_kv`, regex, template mining

AdditionalExtensions, SyslogMessage, RawData

Field inventory

One row per table and field path, with type, fill rate and a sample value. Example rows:

TABLE	FIELD PATH	TYPE	FILL	SAMPLE
CommonSecurityLog	SourceIP	string	0.98	203.0.113.17
AppAudit_CL	Properties.client.addr	dynamic	0.71	198.51.100.4
Syslog	sshd: from <*> port	template	0.64	192.0.2.9

Classify by value, not only by name

Catches badly named columns such as cs4 or field7 that hold IP addresses.

IPv4 / IPv6 · UPN or email · SHA256 / MD5 · URL · Hostname / FQDN

Synonym dictionary

Column-name patterns mapped to ASIM field names. One file.

Template registry

Syslog message templates with each slot mapped to an ASIM field.

Entity index: one narrow table with ASIM column names

Re-run monthly, review the diff

`SourceTable` `SrcIpAddr` `DstIpAddr` `ActorUsername` `DvcHostname` `Url` `FileHash` `Domain`

Sources: Microsoft Learn, data lake KQL and notebooks; LevelBlue design

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